HAMZA AHMED KHAN

AI & GAME DEVELOPER

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PROFESSIONAL OBJECTIVE

Experienced Unity game developer specializing in gameplay mechanics, C# scripting, and optimization, actively expanding technical capabilities into Python, LangChain, and RAG pipelines.

TECHNICAL SKILLS

- **Programming Languages: Python, C#**
- **Frameworks & Tools:** Streamlit, Flask, Git/GitHub, Itch.io, REST APIs, Adobe IllustratorProject
- **Game & Interactive Systems:** Unity (Unity 6), Buildbox, Vuforia (AR), WebGL, Gameplay Logic
- **Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), LangChain, Gemini API, OpenAI API, Vector Search, Prompt Engineering

PROFESSIONAL EXPERIENCE & PROJECTS

Game Developer

Dotlancing, Karachi | Jul 2025 – March 2026

- Designed and built core gameplay mechanics for PC and mobile platforms in fast-paced production cycle
- Optimized rendering loops, script execution, and memory usage to ensure stable framerates across target devices.

Unity Game Developer

Devsall, Karachi | Jul 2024 – May 2025

- *Developed interactive 2D and 3D gameplay systems, managing player controllers, collision mechanics, and state machines*
- *Coordinated on-site with creative teams to test builds, diagnose technical bugs, and refine user feedback loops*

Buildbox Game Developer

Botsolution, Remote | May 2023 – Mar 2024

- *Managed end-to-end development lifecycles for mobile game releases from early prototype to store deployment*
- *Configured core logic, UI layouts, and touch inputs while collaborating asynchronously in an agile setup*

Freelance Game Developer

Self-Employed (Jan 2024 – Present)

- *Developed and delivered custom 2D/3D games, mechanics prototypes, and interactive WebGL experiences for global clients using Unity (C#) and Buildbox.*
- *Engineered modular gameplay systems, resolved complex physics and collision bugs, and optimized draw calls for stable 60 FPS mobile performance.*
- *Integrated third-party SDKs, custom UI/UX assets, and ad networks while managing project milestones and technical requirements from concept to delivery.*

EDUCATION

Bachelor of Science in Computer Science

Federal Urdu University of Arts, Science & Technology, Karachi | 2019– 2023

- Final Year Project: ESENSE — Unity Essence: AR Learning for the Deaf and Mute

FEATURED PROJECTS

AI & LLM PROJECTS

BlogCraft AI — Content & Banner Engine | Python, Gemini API, Streamlit, FLUX

Link: Live Demo | Source Code

- Developed an automated app generating SEO articles and contextual image prompts via the Gemini SDK.
- Integrated Pollinations AI (FLUX) and Streamlit Cloud to synthesize banners and stream content dynamically.

Sentence Craft with Bot — Educational AI Game | Unity, C#, Google Gemini 2.5 Flash, WebGL

Link: Play on WebGL

- Built a 2D WebGL game integrating the Gemini 2.5 Flash API for live sentence and grammar evaluations.
- Created asynchronous C# API routines and CanvasGroup UI states to drive gameplay feedback loops.

GAME DEVELOPMENT PROJECTS

Turn — Lantern Survival Puzzle Platformer | Unity, C#

Link: shhar.itch.io/turn

- Programmed dynamic 2D lighting systems, puzzle collision logic, and custom physics states in Unity.
- Optimized draw calls and memory overhead to ensure stable 60 FPS performance across WebGL browsers.

ESENSE (Unity Essence) — AR Educational App | Unity, Vuforia, C#

- Built an augmented reality tool using Vuforia to assist deaf and mute users through 3D visual gestures.
- Implemented target recognition scripts, camera tracking, and mobile touch interactions in C#.

Phohako Domino — Board Game | Unity, C#

Link: hamza62777.itch.io/phohako-domino

- Programmed turn-based game loop logic, tile-placement validation, and rule-based AI opponent behavior.
- Designed adaptive canvas layouts, custom particle effects, and audio manager scripts.

PORTFOLIO & LINKS

- **LinkedIn:** linkedin.com/in/thehamza62777
- **Game Portfolio (Itch.io): hamza62777.itch.io**
- **Web Portfolio: hamzakhan62777.github.io/PortFolioNew/**

PORTFOLIO
SCAN ME